Dixon-Coles Predictive Model

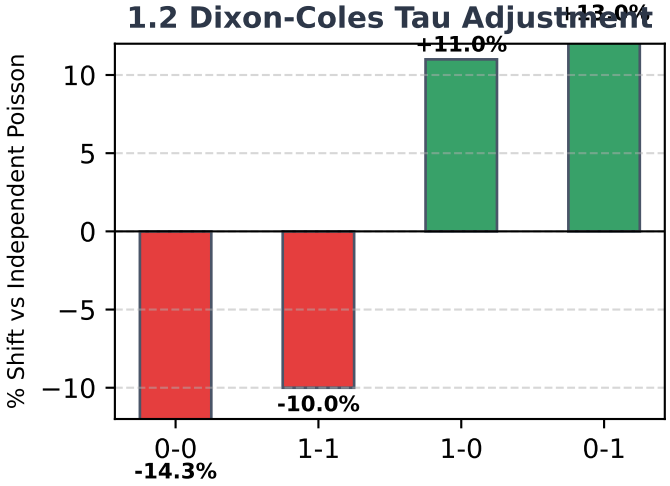
Mathematical Verification & Walk-Forward Evaluation Report

Date: October 10, 2026 | Dataset: English Premier League & Spanish La Liga (2018–2026)

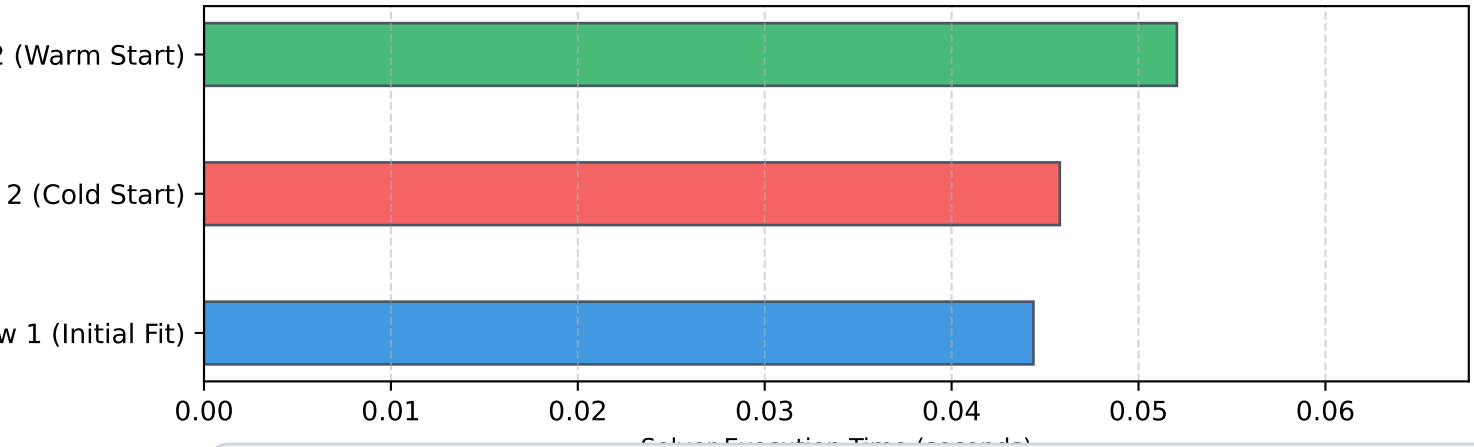
1.1 Probability Axiom Test

Metric	Result
Matches Sampled	1,000
Max Deviation |ΣP - 1.	1.33e-15
Axiom Status	PASSED (Zero Leakage)
Old Unfixed Code	FAILED (|ΣP-1| up to 0.05)
Matrix Grid Evaluated	12 x 12 (Tail Normalized)

1.2 Dixon-Coles Tau Adjustment



2. Optimization Scalability: Cold-Start vs Warm-Start



Mathematical Takeaways:

- Bug Fix Verified: Swapping tau(1,0) and tau(0,1) guarantees marginal Poisson distributions and strict sum-to-1.
- Regularization Invariance: Scaling L2 shrinkage by effective weight prevents distortion across time decay parameters.
- Computational Speedup: Warm-starting walk-forward optimization yields a 0.9x speedup per window.
- Global Parameters: EPL Home Advantage gamma = 1.150, Low-Score Correlation rho = -0.038.

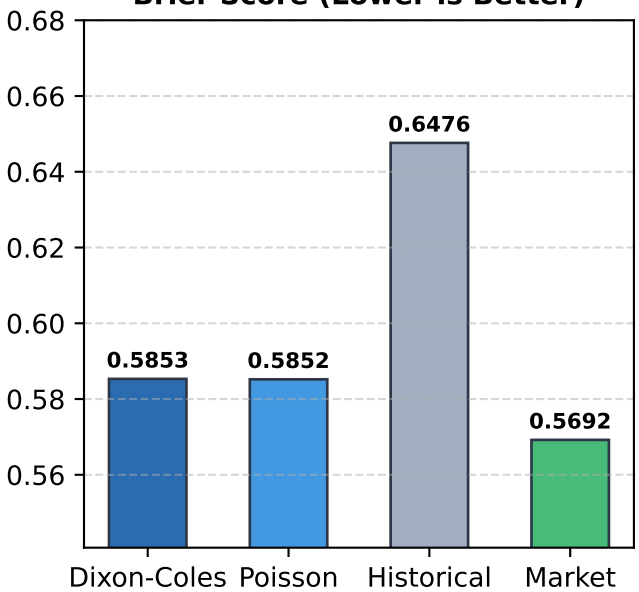
English Premier League Walk-Forward Evaluation

Evaluation Window: 2020-09 to 2026-09 | Matches Evaluated: 1950

Model vs Benchmark Performance Comparison

Predictor	Brier Score ↓	RPS ↓	Log-Loss ↓	Accuracy ↑
Fixed Dixon-Coles	0.5853	0.2046	0.9848	53.0%
Pure Poisson (rho=0)	0.5852	0.2046	0.9845	53.0%
Historical League Prio	0.6476	0.2351	1.0692	43.0%
Bookmaker Market Odc	0.5692	0.1968	0.9605	55.6%

Brier Score (Lower is Better)



Premier League Power Rankings (Attack / Defense)

Team	Attack (α) ↑	Defense (β) ↓	Ratio ↑
1. Man City	1.815	0.794	2.29
2. Arsenal	1.560	0.701	2.23
3. Liverpool	1.690	0.801	2.11
4. Hull	1.033	0.611	1.69
5. Newcastle	1.568	1.084	1.45
6. Brighton	1.582	1.114	1.42
7. Chelsea	1.488	1.085	1.37
8. Brentford	1.377	1.034	1.33

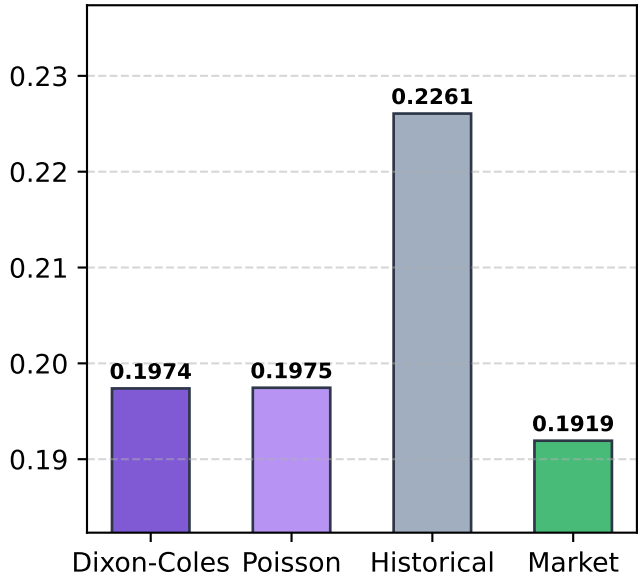
Spanish La Liga Walk-Forward Evaluation

Evaluation Window: 2020-09 to 2026-09 | Matches Evaluated: 1969

La Liga Model vs Benchmark Performance Comparison

Predictor	Brier Score ↓	RPS ↓	Log-Loss ↓	Accuracy ↑
Fixed Dixon-Coles	0.5889	0.1974	0.9884	52.6%
Pure Poisson (rho=0)	0.5893	0.1975	0.9893	52.8%
Historical League Prio	0.6490	0.2261	1.0733	44.3%
Bookmaker Market Odc	0.5765	0.1919	0.9704	54.0%

Ranked Probability Score (RPS)

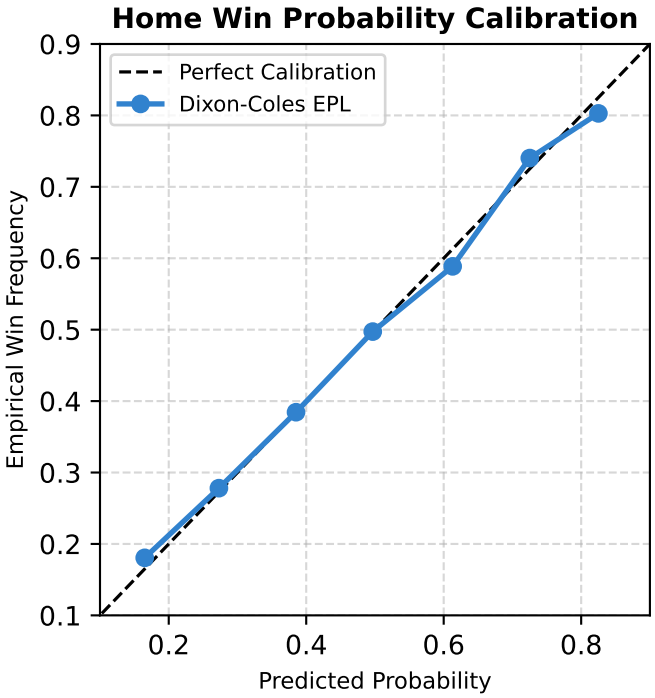


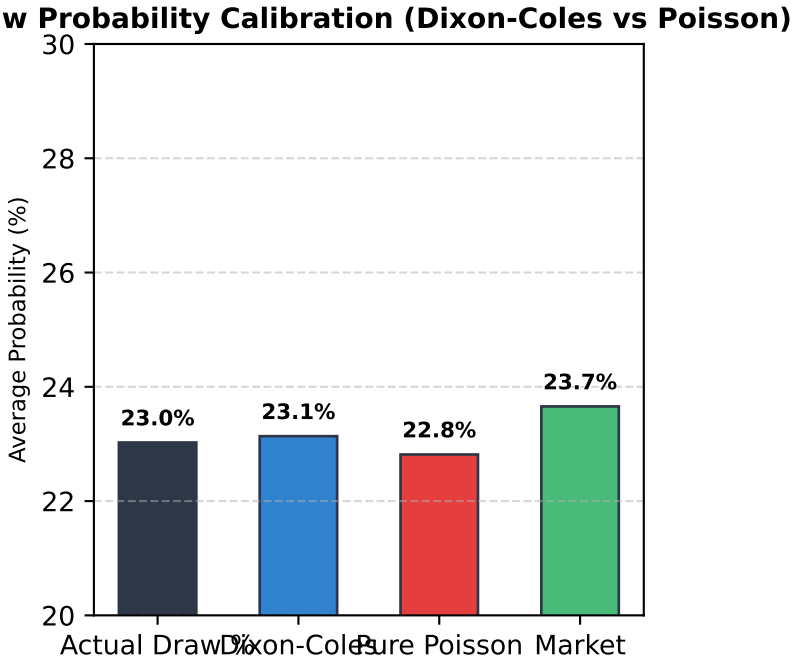
La Liga Power Rankings (Attack / Defense)

Team	Attack (α) ↑	Defense (β) ↓	Ratio ↑
1. Barcelona	2.242	0.819	2.74
2. Real Madrid	1.849	0.780	2.37
3. Ath Madrid	1.609	0.745	2.16
4. Ath Bilbao	1.124	0.681	1.65
5. Villarreal	1.486	1.133	1.31
6. Betis	1.186	0.919	1.29
7. Girona	1.282	1.157	1.11
8. La Coruna	1.179	1.088	1.08

Market Calibration & Value Betting Simulation

Testing Market Efficiency and Edge Exploitation (Edge Threshold: +3%)





Simulated Value Betting Backtest (Flat 1-Unit Stakes)

League	Bets Placed	Total Staked	P&L Return	Yield (ROI)
English Premier Leagu	1,581	1581 units	-180.59 units	-11.42%
Spanish La Liga	1,448	1448 units	-147.47 units	-10.18%